Alcohol Hangover and Driving-Relevant Cognition: A Dual-Task & Modelling Approach

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Background

- Alcohol Hangover = Symptoms after heavy alcohol use when BAC ≈ 0%.
- Impairs cognition, motor function, and driving safety (Verster et al., 2020; Gunn et al., 2018).
- Unlike intoxication, no roadside test for detection.
- Cognitive deficits from hangover → underestimated public health risk.

Research Gaps Addressed

- ✕ Limited application of advanced cognitive models.
- ✕ Previous studies lacked detailed decomposition of decision-making processes.
- ✕ Disconnect between subjective hangover symptoms and objective cognitive performance measures.
- ✕ Didn’t examine variability in temporal estimation.

- ✓ Applied Hierarchical Drift Diffusion Modelling (HDDM) to break down cognitive components under hangover state.
- ✓ Combined subjective self-reports with objective cognitive performance measures.
- ✓ Introduced a novel anticipatory motor timing task to assess temporal estimation variability.

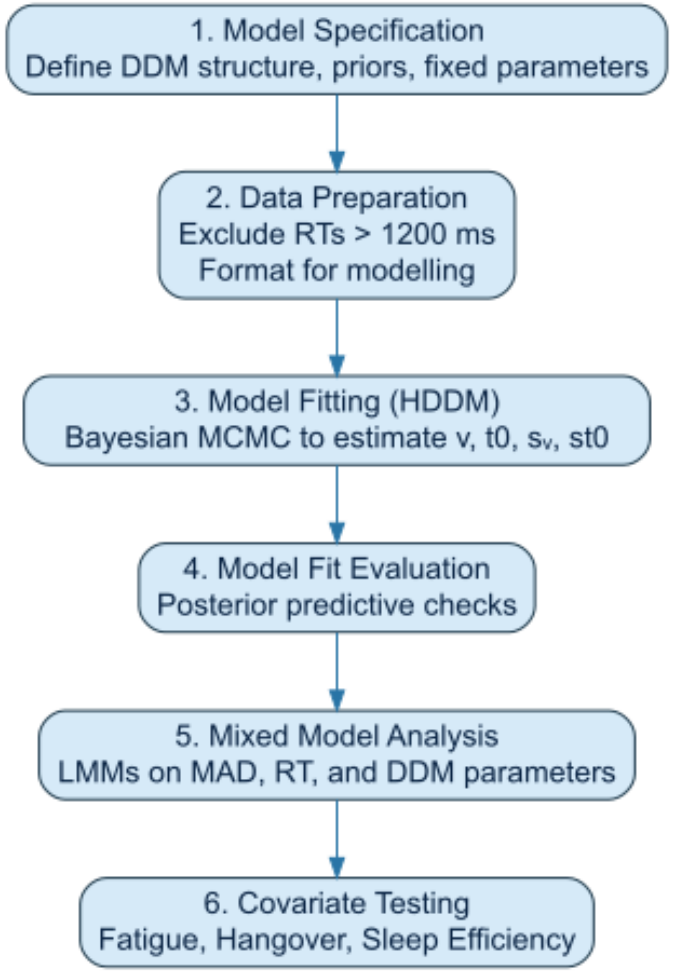
Study Aims

- Assess psychomotor speed (using reaction time).
- Model decision-making processes using Hierarchical Drift Diffusion Modelling (HDDM).
- Quantify temporal estimation variability in a novel motor timing task.
- Explore the influence of self-reported hangover severity, fatigue, and sleep ratings.

Hypotheses

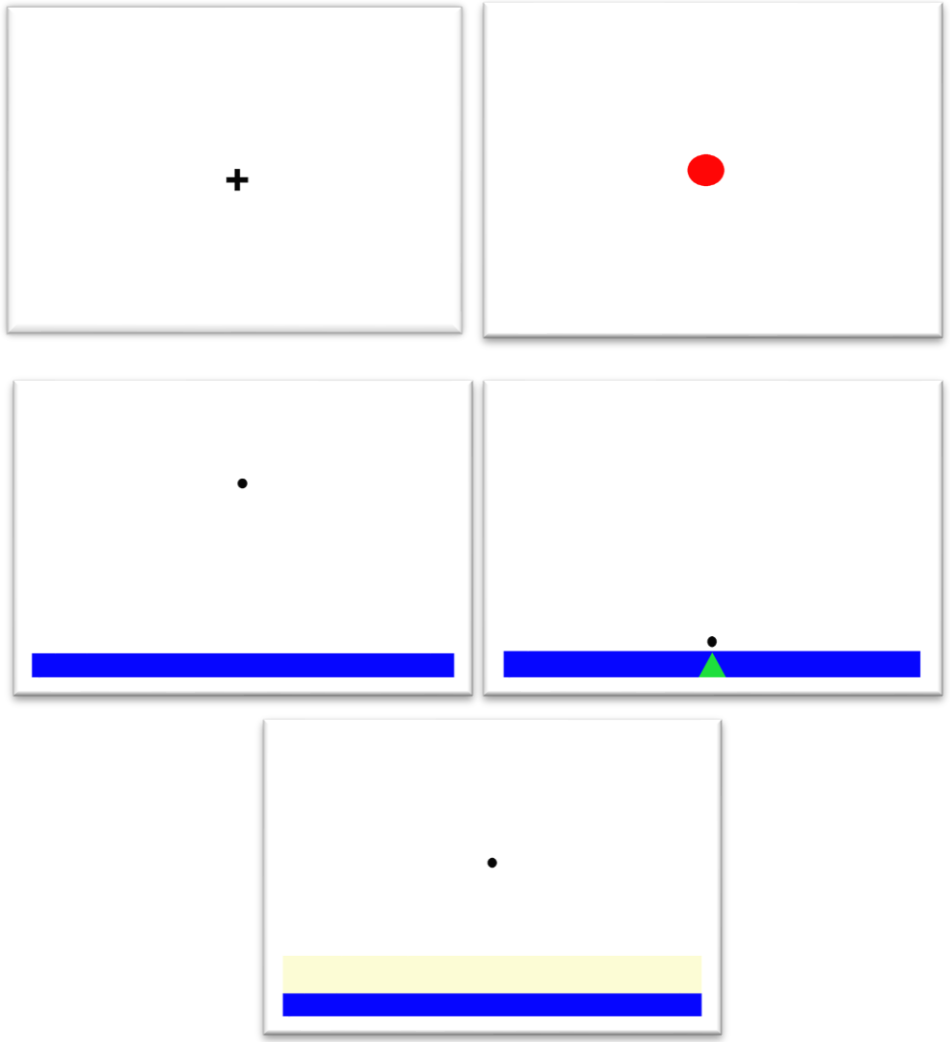
- **H1** (slower psychomotor speed): Longer RTs in the simple reaction time task.
- **H2** (DDM): Reduced drift rate (v) and increased (sv, st_0).
- **H3** (temporal estimation): Greater variability (higher MAD) in the anticipatory timing task.
- **H4** (covariate analysis): Fatigue, sleep and hangover severity would be significant predictors of task performance.

Research Design



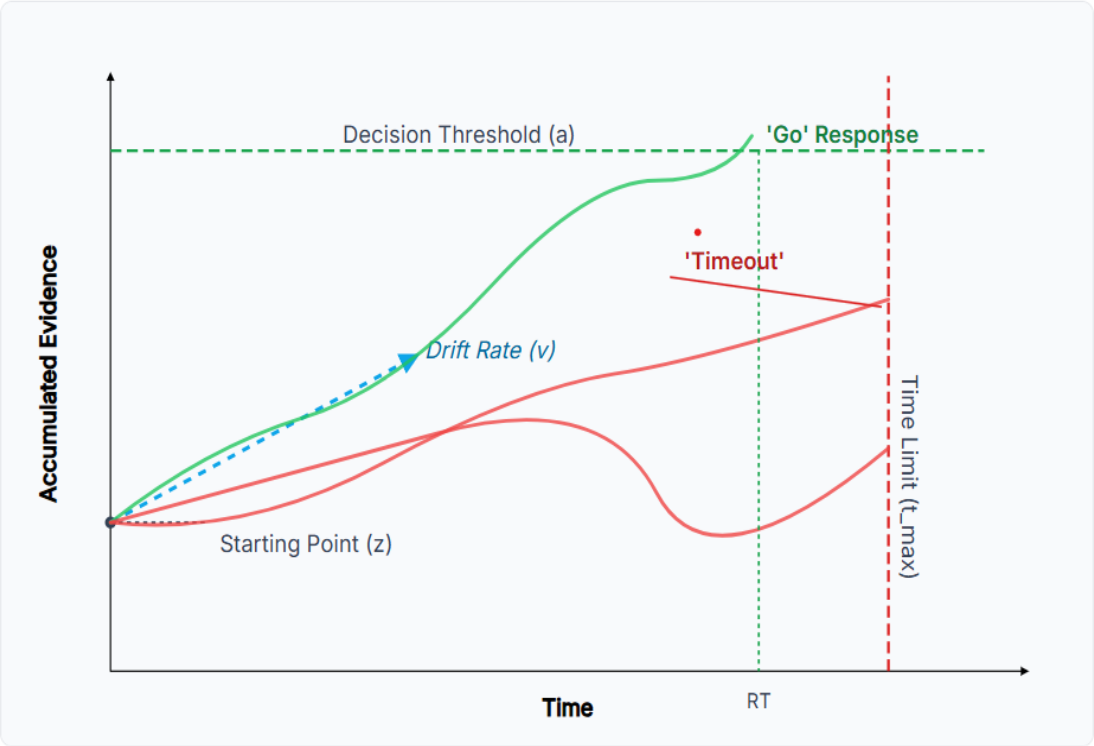
- Within-subjects, randomised crossover design
- **N = 19** undergraduates from the UWA School of Psychology with a valid driver's license.
- Procedure:
  - Hangover: Naturalistic drinking the night before, testing 8-10 hrs, post-consumption.
  - Control: Abstinence >24 hrs.
  - Testing session involved cognitive tasks and questionnaires.
  - Hangover and sober conditions counterbalanced.

Computerised Cognitive Tasks



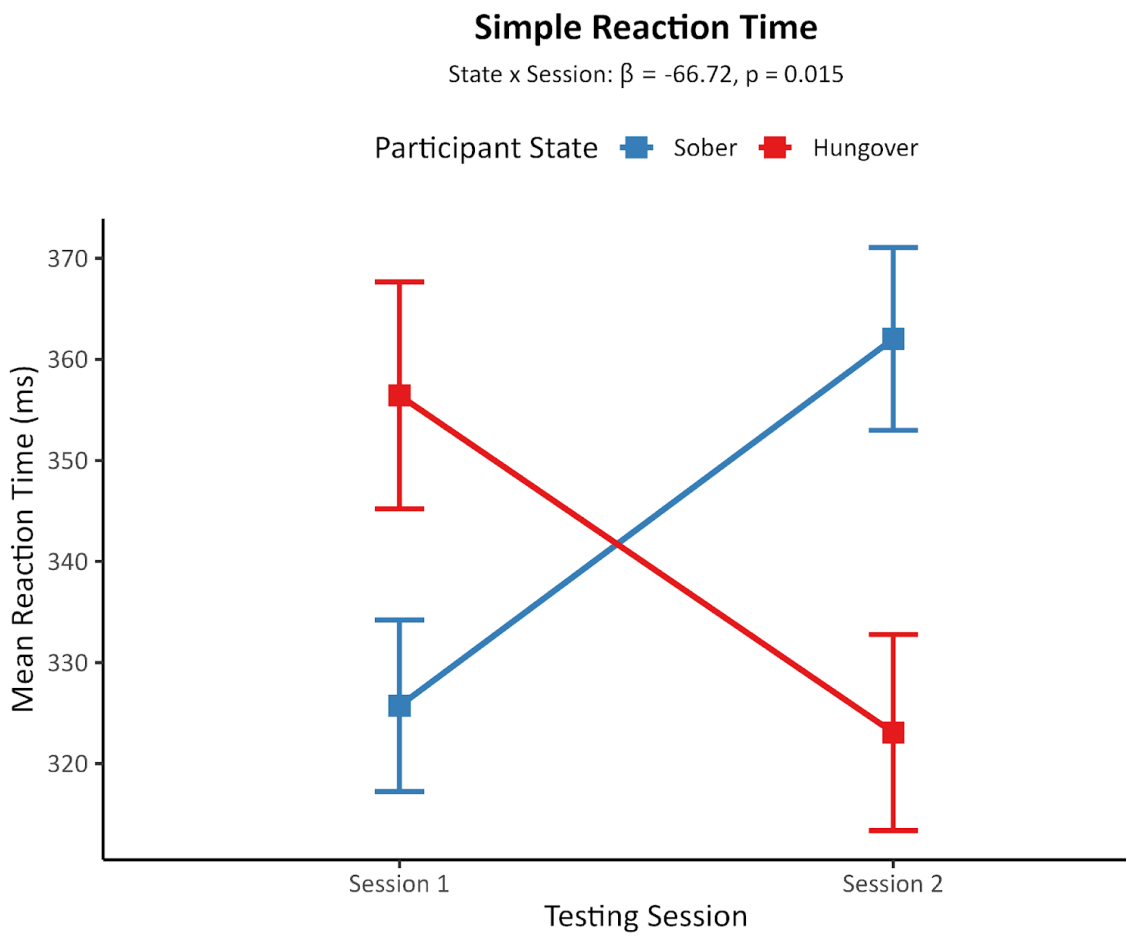
- Simple Reaction Time Task – Press spacebar when red circle appears.
- Anticipatory Timing Task – Predict timing of a falling disk across:
  - Baseline (visible trajectory)
  - Distracted (irrelevant visual cue)
  - Occluded (partial trajectory hidden)
  - Speed Variation (random speed changes)

Modelling Approach



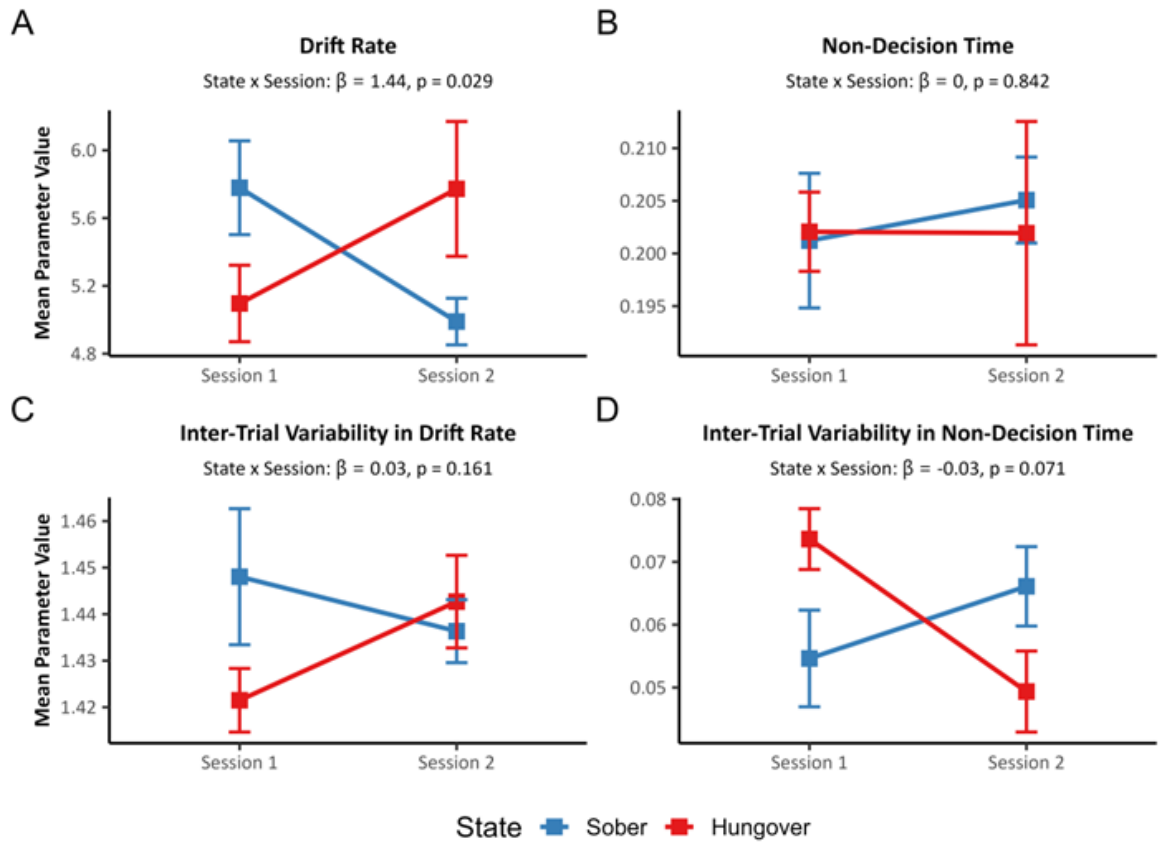
- HDDM Parameters:
  - Drift Rate (v) – Evidence accumulation speed.
  - Non-Decision Time (t_o) – Motor/perceptual processes.
  - Inter-trial variability (sv, st_o) – Performance fluctuations.
- Implemented via Stan Bayesian framework with MCMC sampling.

Key Results: Psychomotor Speed



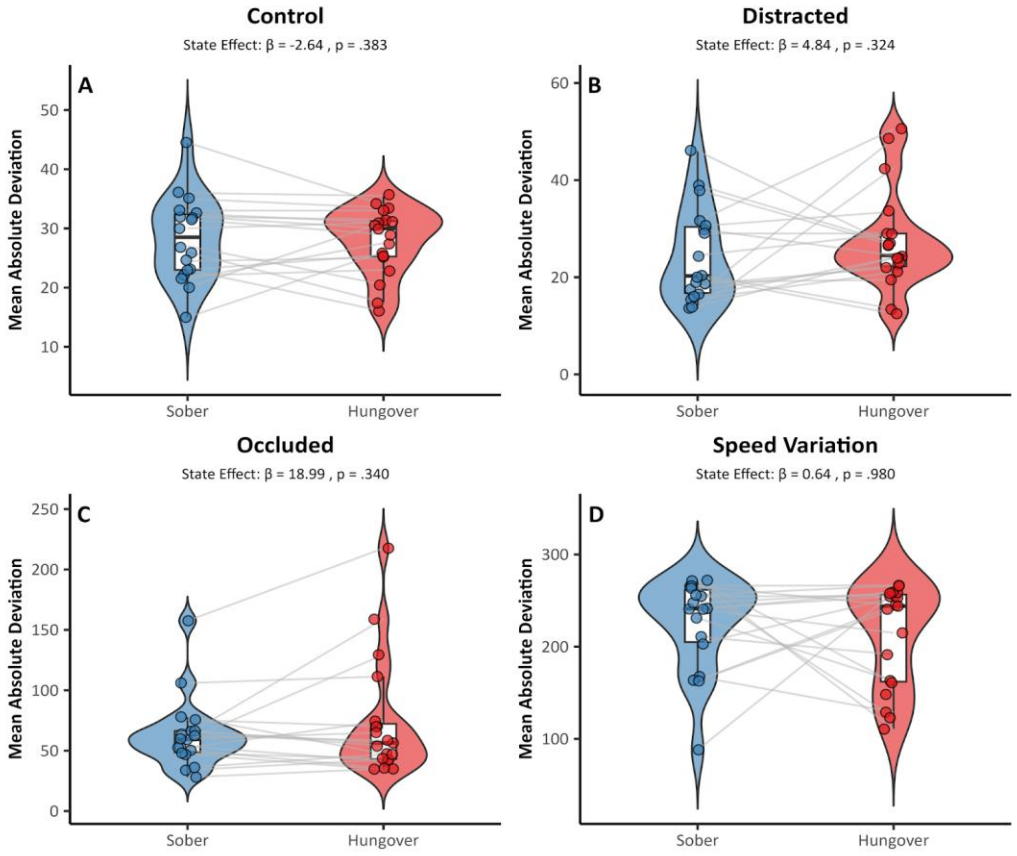
- Significant state x session interaction for RT (p = .015).
- RT slower in hangover (p = .042).
- Learning effects complicate interpretation.

Decision-Making Modelling Results



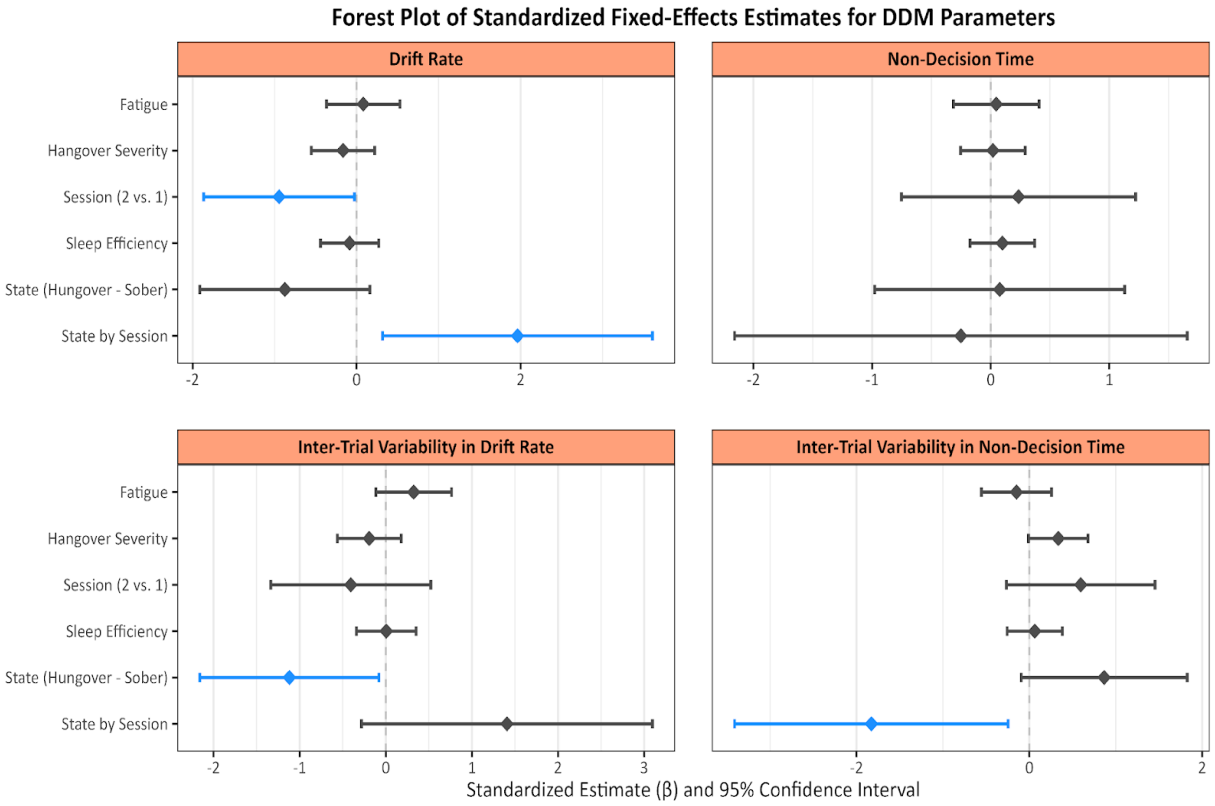
- Drift Rate (*v*): Interaction effect (*p* = .029), main effect non-significant.
- Inter-trial variability:
  - *sv* lower in hangover (*p* = .049).
  - st_0 higher in hangover (*p* = .044).
- Suggests noisier cognitive and motor processes in hangover

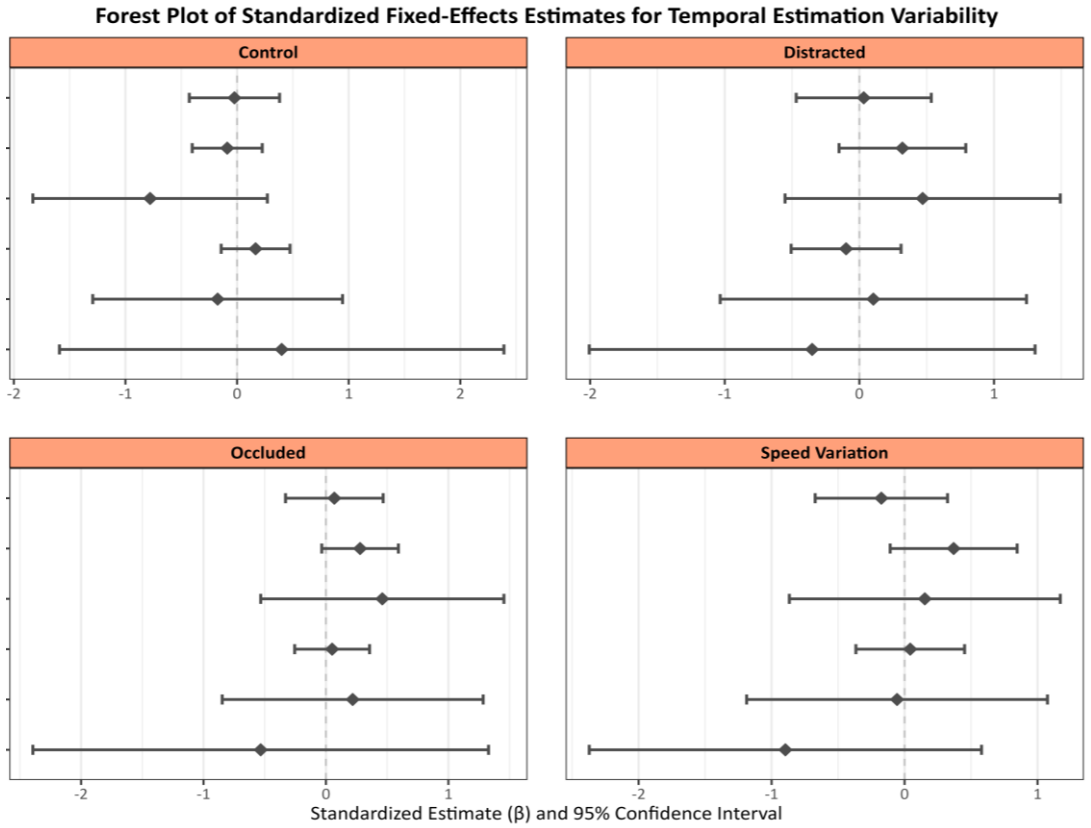
Temporal Estimation Variability



- Assessed temporal estimation variability using Mean Absolute Deviation (MAD).
- No significant main effect of hangover state.
- Directionally higher MAD in hangover for Occluded, but wide confidence intervals.
- Substantial individual variability in performance.

Covariate Analysis





Conclusions & Implications

- Mild hangover impairs psychomotor speed and decision-making stability.
- Temporal estimation less affected – suggests domain-specific vulnerability.
- Self-perception (fatigue, sleep) does not always align with objective impairment.
- Need for:
  - Studies across broader hangover severities.
  - Real-world driving simulations.
  - Integration with neuroimaging & wearable monitoring.

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